



Draw the root locus for the following transfer using the rules.

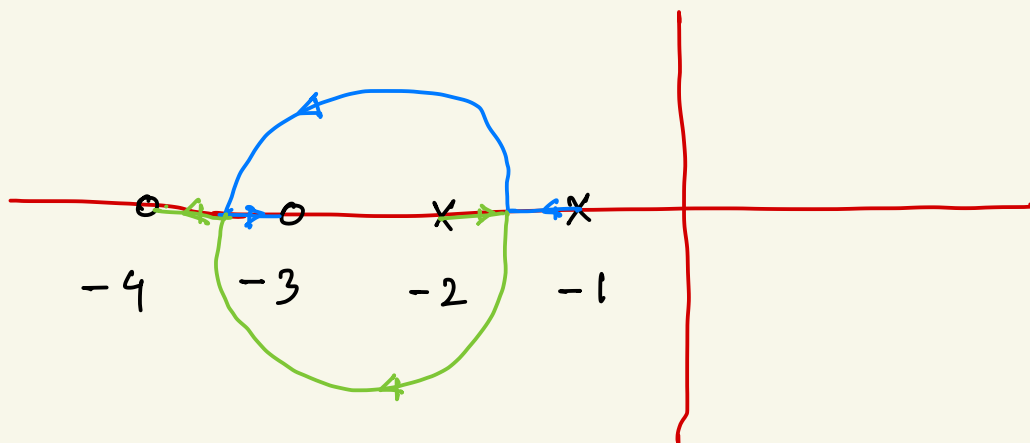
$$G(s) = K \frac{(s+3)(s+4)}{(s+1)(s+2)}$$

number of poles $n = 2$

number of zeros $m = 2$

number of asymptotes $n - m = 0$

root locus starts at poles and ends at zeros.



A case study :

Design an autopilot so the response to a step elevator input has a rise time of 1 sec or less and an overshoot less than 10%.

The transfer function to a Piper Dakota aircraft is as follows

$$G(s) = \frac{\theta(s)}{\delta_e(s)} = \frac{160(s+2.5)(s+0.7)}{(s^2+5s+40)(s^2+0.03s+0.01)}$$

θ = pitch attitude

δ_e = elevator angle in degrees.

Ans:- Rise time t_r can be approximated by

$$t_r = \frac{1.8}{\omega_n}$$

$$\Rightarrow 1 = \frac{1.8}{\omega_n}$$

$$\Rightarrow \omega_n = 1.8 \text{ Hz}$$

$$M_p = e^{-\zeta\pi/\sqrt{1-\zeta^2}} = 0.1$$

$$\Rightarrow \pi\zeta/\sqrt{1-\zeta^2} = \ln 0.1$$

$$\Rightarrow \frac{\pi^2\zeta^2}{(1-\zeta^2)} = (\ln 0.1)^2$$

$$\Rightarrow \pi^2\zeta^2 = (\ln 0.1)^2 - \zeta^2(\ln 0.1)^2$$

$$\Rightarrow \pi^2\zeta^2 + \zeta^2(\ln 0.1)^2 = (\ln 0.1)^2$$

$$\text{Simplifying } \xi = \sqrt{\frac{(\ln 0.1)^2}{\pi^2 + (\ln 0.1)^2}}$$

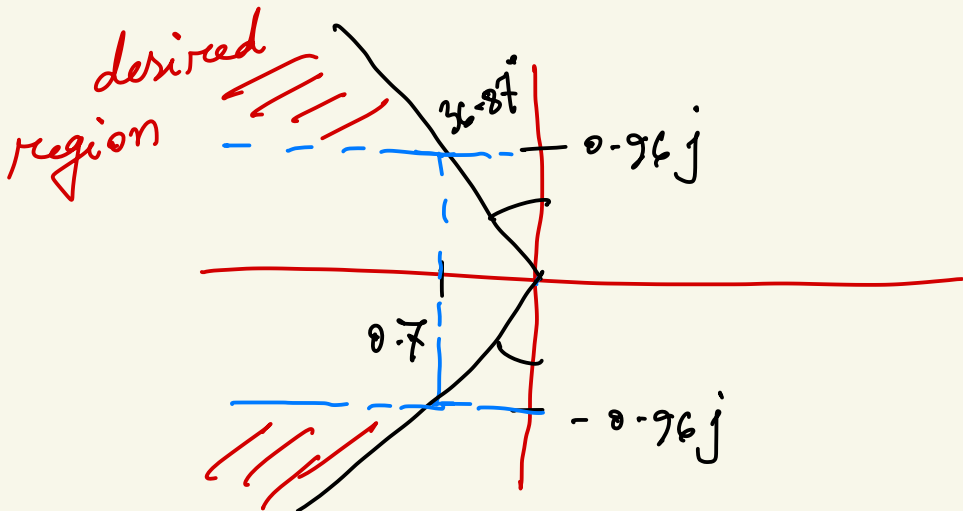
$$= 0.5912$$

That is ξ needs to be 0.6 or greater.

$$\theta = \sin^{-1} \xi = 36.87^\circ$$

$$\sigma = -\xi \omega_n = 0.5912 \times 1.2 = 0.7$$

$$\omega_d = \omega_n \sqrt{1 - \xi^2} = 0.96$$



The transfer function is a fourth order system. We need to draw the root locus in Matlab and check how the poles are moving and how can we get the dominant poles in the desired region.

Matlab code uploaded and details are discussed.

Compensators :-

Compensators are a kind of controller that can shift the position of the root locus to a

desired position.

Usually, there are two basic type of compensator.

1. Lead compensator
2. Lag compensator

A general form of the compensation is as follows:

$$D_c(s) = K \frac{s + z}{s + p}$$

If $p > z$, it is called a lead compensator.

If $z > p$, it is called a lag compensator.

A lead compensator approximates the function of a PD controller. It can lower the rise time and decrease overshoot.

On the other hand, a lag compensation approximates PI control and improve steady-state accuracy.

In this case study problem, a lead compensator was used

to tweak the root locus and poles into desired locations.

check Matlab code.